

Graduate Research Plan Statement

Geospatial machine learning (GeoML) faces an inevitable challenge: the sparsity of labeled training data. The archive of unlabeled geospatial data (e.g., satellite imagery) is large and growing, which makes it intractable to obtain labels or measurements in all geographic regions. GeoML models are frequently used to make predictions in places lacking labeled data, but to do so accurately, these models rely on the quality of initial labeled training data. Developing a training set often requires human input as to where and when to sample the data; these decisions are often difficult, due to resource constraints, and subjective. To generate high-quality training data for GeoML models, I propose ML-informed data collection through an optimal dataset design framework.

Background: Currently, active learning and adaptive sampling techniques seek to optimize ML model performance by iteratively sampling data on which the model is most uncertain^{1,2}. Tuia et al.¹ presents two adaptive sampling techniques for remote sensing image classification, both of which use measures of model uncertainty. However, when the initial dataset is small, points from certain subgroups may not be present in this initial data. Existing sampling methods will overlook these points if they are correctly classified by the model, which may result in a training set that is not representative of these subgroups. Research outside of GeoML has demonstrated the effect of subgroup representation on learning outcomes³. In the geospatial context, this work motivates the representation of spatially defined groups (e.g., groups representing different countries or regions) in the initial dataset. **I propose an optimal dataset design framework that leverages both model uncertainty and spatial representation.** To be useful to scientists allocating resources for data collection, **this framework will assess the feasibility of labeling a dataset based on resource constraints**—an approach currently not addressed in the GeoML literature.

Research Plan: For a given GeoML model, what training points can and should be labeled to optimize model performance? Formally, let f_θ represent the model trained on X_{train} ; the proposed work will solve:

$$\operatorname{argmax}_{X_{train}} \operatorname{Accuracy}(f_\theta(X_{test})) \text{ s.t. } X_{train} \text{ is feasible to collect.}$$

The solution to this problem involves two components:

- A sampling scheme: a procedure that outputs a set X_{train} that maximizes $\operatorname{Accuracy}(f_\theta(X_{test}))$
- A feasibility function: a binary function that informs the feasibility of collecting X_{train}

To develop these components effectively, I will first understand factors that influence optimal dataset design for spatial problems by (i) experimentation on existing datasets and (ii) implementation for an unseen dataset. This will inform (iii) the development of an optimal dataset design framework.

Aim 1: Understanding factors of dataset composition through experimentation on existing datasets.

Geospatial data can be modeled as a spatial random process (SRP)—a set of random variables indexed over a domain of locations L . Given an estimator over L , we can determine a subset of L that minimizes the mean squared error of the estimator's predictions. I will implement a sampling scheme [called SS_1] which selects a set X_{train} using this spatial information, provided by the solution to the SRP problem; SS_1 serves as a **spatial-only baseline**. I will also implement a sampling scheme [called SS_2] that uses a measure of model uncertainty (e.g., an entropy-based heuristic for classification tasks) to select a set X_{train} ; SS_2 will serve as an **uncertainty-motivated baseline**. To understand which factors of dataset composition maximize $\operatorname{Accuracy}(f_\theta(X_{test}))$, I will leverage both spatial information and model uncertainty in the third sampling scheme, SS_3 . Specifically, SS_3 will output a sample from SS_1 with probability λ and a sample from SS_2 with probability $1 - \lambda$. I hypothesize that there exists a $0 < \lambda < 1$ for which SS_3 will output a sample X_{train} that yields better model performance than SS_1 and SS_2 alone. Evaluation for this stage will involve experimentation on 3 commonly studied datasets in GeoML: EuroSAT^{4,5}, Kenya Crop Type⁶, and USAVars⁷. To simulate feasibility constraints in these existing datasets, I will devise a feasibility function that compares travel cost for a set X_{train} to a budget, k . I will then implement SS_1 , SS_2 and SS_3 , which will output a subset of the dataset that satisfies this feasibility function. Success will be evaluated by the comparison of model accuracy after training on X_{train} from SS_1 , SS_2 and SS_3 . **Challenges:** Solving the SRP problem involves understanding the covariance matrix of the random variable at locations L . In practice, if the covariance matrix is not well-understood, then I will have to impose assumptions on the form of the

covariance to solve the problem. If such assumptions yield unrealistic results, I will also try a variant of SS_1 that involves grid-based spatial sampling. Additionally, the feasibility function is grounded in real-world constraints but may not be adequate in certain contexts. This will be addressed in the next two aims.

Aim 2: Implementation for an unseen dataset. This dataset design framework will be particularly useful for two collaborators of the Rolf Lab, namely an ecologist and an environmental scientist, who collect training data for GeoML models across several geographic regions. First, the feasibility function will be adjusted to represent each collaborator's resource and budget constraints. Based on the intended task, I will demonstrate that the proposed sampling scheme SS_3 (with respect to the feasibility function) yields higher model accuracy than pre-existing data sampling methods. Following this demonstration, I will implement SS_3 for each collaborator's intended model and task, specified by my collaborator. Success for this portion of the project will be measured by feedback from collaborators, particularly involving the ease of adoption.

Challenges: With various opinions on the best way of sampling, it may be challenging to convince the necessary personnel of the power of ML-informed data collection. However, the two collaborators of the Rolf Lab will advocate for implementation of this framework to adequately allocate data labeling resources.

Aim 3: Using these results to inform optimal dataset design. With results on existing datasets (Aim 1) and on an unseen dataset (Aim 2), I will **critically re-evaluate** the factors of dataset composition in the proposed sampling scheme, SS_3 and the validity of the feasibility function. In this stage, I will

- Quantify subgroup performance of f_θ for relevant subgroups (e.g., biomes, ecoregions, elevation levels). This will determine if X_{train} , selected by SS_3 , is representative of these subgroups.
- Determine if the feasibility function accommodates necessary user specifications

Following these assessments, I will **1.** extend SS_3 to select samples from each relevant subgroup with a specified probability [called SS_4], and **2.** generalize the feasibility function to accommodate variables highlighted in Aim 2, including a confidence measure to address constraints that may be unaccounted for. Success will be evaluated by experimentation on the same datasets from Aim 1. After separately training models on sets selected from SS_3 and SS_4 , I will compare model performance on the entire dataset *and* model performance on relevant subgroups of that dataset. **Challenges:** To quantify subgroup performance, I will need to decide which subgroups of each dataset are relevant. With this subjectivity, it is possible that important subgroups may be overlooked. To report my findings, I will specify that these results represent a comparison of subgroup performance for a fixed subgroup, and not necessarily for all subgroups.

Intellectual Merit

This project aims to advance knowledge in the field of geospatial machine learning by first understanding the impact of spatial representation on model performance (Aim 1). Previous work highlights the impact of subgroup representation on both subgroup and global performance for non-geospatial data³. By leveraging spatial representation, the proposed sampling scheme prompts the GeoML model to **perform well both on a global scale and on a local scale**. Furthermore, by introducing a feasibility function to the GeoML literature, the proposed framework works toward **realistic implementation for domain scientists** (Aim 2). Lastly, Aim 3 will offer a thorough analysis of the proposed framework, **encouraging future work on ML-informed optimization of data collection**.

Broader Impacts

Domain scientists frequently travel to remote and, often, unsafe locations for data collection and census efforts. This framework will **empower these scientists to make informed decisions about data collection**, while understanding both the implications for GeoML model performance and the feasibility of collecting these data points. Once collected, this data serves as training data for GeoML models, which are increasingly used to inform public policy decisions. For example, following the COVID-19 pandemic, governments used a GeoML model to decide where to allocate humanitarian aid⁹. This project takes **critical and achievable steps toward making these models more accurate for public institutions**.

References: [1] Tuia et al. (2009) IEEE [2] Aiken et al. (2023) IJCAI [3] Rolf et al. (2021) PMLR [4] Helber et al. (2018) IEEE [5] Helber et al. (2019) IEEE [6] CV4A Comp. (2020) [7] Rolf et al. (2020) CoRR [8] Stewart et al. (2021) CoRR [9] Aiken et al. (2022) Nature